CBSE Class 10 Mathematics - Answer Key

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1. Which of the following is a quadratic equation?

Answer: (C)

Solution:

1. Expand each option to check for the highest power of x.
2. A quadratic equation must be in the form $ax^2 + bx + c = 0$ where $a \neq 0$.
3. Option (C) results in $x^2 + 3x + 1 = x^2 - 4x + 4$, which simplifies to $7x - 3 = 0$ (Linear). Wait, let's recheck.
4. Actually, checking $x^2 - 2x = (-2)(3 - x) \Rightarrow x^2 - 2x = -6 + 2x \Rightarrow x^2 - 4x + 6 = 0$.

Derivation:

$$\begin{aligned} x^2 - 2x &= -6 + 2x \\ \Rightarrow x^2 - 4x + 6 &= 0 \end{aligned}$$

2. The discriminant of the quadratic equation $2x^2 - 4x + 3 = 0$ is

Answer: (B)

Solution:

1. Identify $a = 2, b = -4, c = 3$.
2. Use the formula $D = b^2 - 4ac$.
3. $D = (-4)^2 - 4(2)(3) = 16 - 24 = -8$.

Derivation:

$$D = (-4)^2 - 4 \times 2 \times 3 = 16 - 24 = -8$$

3. If $x = 1/2$ is a root of the equation $x^2 + kx - 5/4 = 0$, then the value of k is

Answer: (A)

Solution:

1. Substitute $x = 1/2$ into the equation.
2. $(1/2)^2 + k(1/2) - 5/4 = 0$.
3. $1/4 + k/2 - 5/4 = 0 \Rightarrow k/2 - 1 = 0 \Rightarrow k = 2$.

Derivation:

$$\begin{aligned} 1/4 + k/2 - 5/4 &= 0 \\ \Rightarrow k/2 &= 4/4 \\ \Rightarrow k/2 &= 1 \\ \Rightarrow k &= 2 \end{aligned}$$

4. The quadratic equation $x^2 + 4x + k = 0$ has real and equal roots if

Answer: (C)

Solution:

1. For real and equal roots, $D = 0$.
2. $b^2 - 4ac = 0$.
3. $4^2 - 4(1)(k) = 0 \Rightarrow 16 - 4k = 0 \Rightarrow k = 4$.

Derivation:

$$\begin{aligned} 16 - 4k &= 0 \\ \Rightarrow 4k &= 16 \\ \Rightarrow k &= 4 \end{aligned}$$

5. If the equation $x^2 + 5kx + 16 = 0$ has no real roots, then

Answer: (D)

Solution:

1. For no real roots, $D < 0$.
2. $b^2 - 4ac < 0$.
3. $(5k)^2 - 4(1)(16) < 0 \Rightarrow 25k^2 - 64 < 0 \Rightarrow k^2 < 64/25 \Rightarrow -8/5 < k < 8/5$.

Derivation:

$$\begin{aligned} 25k^2 &< 64 \\ \Rightarrow k^2 &< 64/25 \\ \Rightarrow |k| &< 8/5 \end{aligned}$$

6. The roots of the equation $x^2 - 3x - 10 = 0$ are

Answer: (B)

Solution:

1. Factorise $x^2 - 3x - 10 = 0$.
2. $x^2 - 5x + 2x - 10 = 0$.
3. $x(x - 5) + 2(x - 5) = 0 \Rightarrow (x - 5)(x + 2) = 0$.

Derivation:

$$x = 5, x = -2$$

7. Which constant must be added to $x^2 + 6x$ to make it a perfect square?

Answer: (B)

Solution:

1. The coefficient of x is 6.

2. Half of the coefficient is 3.
3. Square of 3 is 9.

Derivation:

$$(6/2)^2 = 3^2 = 9$$

8. The sum of the roots of the quadratic equation $3x^2 - 5x + 2 = 0$ is

Answer: (C)

Solution:

1. For $ax^2 + bx + c = 0$, sum of roots $= -b/a$.
2. Here $a = 3, b = -5$.
3. Sum $= -(-5)/3 = 5/3$.

Derivation:

$$Sum = -b/a = -(-5)/3 = 5/3$$

9. If one root of $x^2 - 8x + m = 0$ is 3, then the value of m is

Answer: (D)

Solution:

1. Substitute $x = 3$ into the equation.
2. $3^2 - 8(3) + m = 0$.
3. $9 - 24 + m = 0 \Rightarrow -15 + m = 0 \Rightarrow m = 15$.

Derivation:

$$\begin{aligned} 9 - 24 + m &= 0 \\ \Rightarrow m &= 15 \end{aligned}$$

10. The equation $(x+1)^2 - x^2 = 0$ has how many real roots?

Answer: (A)

Solution:

1. Expand $(x+1)^2 - x^2 = 0$.
2. $x^2 + 2x + 1 - x^2 = 0$.
3. $2x + 1 = 0 \Rightarrow x = -1/2$.
4. This is a linear equation, so it has exactly one root.

Derivation:

$$\begin{aligned} 2x + 1 &= 0 \\ \Rightarrow x &= -0.5 \end{aligned}$$